

Rishit Yogesh Bafna

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EDUCATION

- Arizona State University**, *M.S. Computer Science* – GPA: 3.44 Tempe, AZ | Jan 2026 – May 2027
- Arizona State University**, *B.S. Computer Science (SWE)* – GPA: 3.42 Tempe, AZ | Earned Dec 2025
- Dean's List (multiple semesters)
 - Relevant Coursework: Operating Systems, Distributed Systems, Machine Learning, Data Structures and Algorithms, Software Design

PROJECTS

- F1 Race Strategy RL** (*Ongoing*) | *Python, PyTorch, PPO, FastAPI, React*
- Built a multi-agent PPO reinforcement learning environment from scratch over 188K laps across 172 F1 races (2018-2025).
 - Trained three physics sub-models for tire degradation, safety car prediction, and DNF risk, reaching 0.81 AUC on safety car classification.
 - Designed a 142-dimension observation space with counterfactual reward shaping across all 20 cars per race.
- RiskStream** (*May 2026*) | *Python, XGBoost, FastAPI, PostgreSQL, Redis, MLflow, Docker*
- Engineered 23 behavioral features, including 7-day personalized spending baselines, with zero database reads on the inference hot path.
 - Achieved 0.94 PR-AUC at 0.17% class imbalance across four simulated fraud attack patterns, with p99 inference latency under 50ms.
 - Automated PSI-based drift detection and model hot-swapping in production, eliminating manual retraining and service restarts.
- CyberSentient RAG Pipeline** (*May 2026*) | *Python, Qwen3-Embedding-8B, Qdrant, BM25, FastAPI*
- Built hybrid retrieval over 100K+ CVE/CWE/CAPEC records by combining dense vector search, BM25, and Reciprocal Rank Fusion.
 - Reduced false retrievals on ambiguous CVE lookups using Qwen3-Reranker-8B cross-encoder reranking.
 - Achieved 400ms end-to-end query latency across a 6-feed real-time threat intelligence pipeline.
- Distributed Key-Value Store** (*Jan 2026*) | *Java, Raft Consensus, gRPC, Protocol Buffers*
- Implemented a fault-tolerant key-value store using Raft consensus across a 5-node cluster.
 - Built a gRPC API for SET/GET/DELETE with correct leader redirection at sub-5ms latency.
 - Verified zero data loss and automatic recovery through chaos testing of simulated leader failures.

EXPERIENCE

- Software Engineering Intern**, *J. Miller Custom Cues* Tempe, AZ | Aug 2025 – Dec 2025
- Shipped production 3D product configurator using Three.js and React, reducing customer design revision cycles by 35%.
 - Built PostgreSQL REST API with Stripe integration processing 100+ orders with transactional cart management and sub-200ms response times.
- Software Engineering Intern**, *Winssoft Technologies India Pvt. Ltd.* Mumbai, India | May 2025 – Jul 2025
- Optimized SQL queries over 200K+ records using composite indexes and materialized views, reducing dashboard load time from 8s to 2s.
 - Refactored payment microservice handling 5K+ daily transactions with idempotency guarantees, preventing duplicate charge errors in concurrent request scenarios.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, SQL
Backend: FastAPI, Node.js, Express.js, REST APIs, gRPC, WebSockets
Databases & Tools: PostgreSQL, Redis, MongoDB, MLflow, Docker, Git, Linux, AWS (S3, EC2)
Machine Learning: XGBoost, scikit-learn, RAG Pipelines, Vector Databases (Qdrant)
Frontend: React, Three.js, Next.js, Tailwind CSS